BRAC University | CSE-460: VLSI Design | Spring 26 | Faculty: DPRN

Assignment - 03

Due - 11:59 PM, April 23, 2025 | Submission link

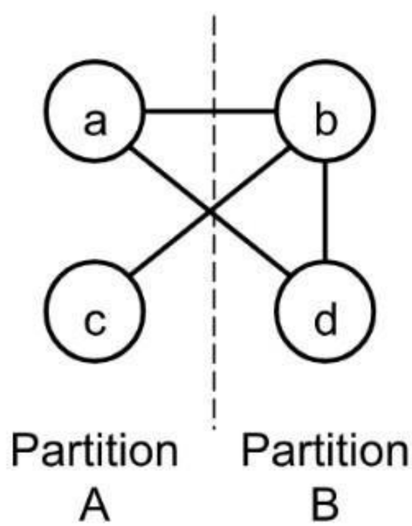
ID:	Sec:	Name:

Instructions:

- **Print the question paper** and start writing your answers as instructed.
- **Do not print the questions on both sides of the pages**. You can **use the opposite side of the page** for writing your answers.
- You can add additional pages if required, but you **must write your ID, Section and Name at the top of each additional page.**
- Please note, **Collaboration =/= copying.** You are allowed to discuss the questions and clear confusions you might have, but you have to write your solutions independently and distinctively.
- **Scan** your answers, make a single PDF file by renaming "A3_StudentID_Fullname_CSE460.pdf" and **upload** it in the submission link.
- **Any findings of plagiarism will** result in a **Zero** mark. It may also hamper your future grading!
- **You must submit the online copy by the deadline.**

Question-1

The graph below (nodes a-d) can be optimally partitioned using the Kernighan-Lin algorithm. The dotted line represents the initial partitioning. Assume all the edges have the same weight.



a. Calculate the initial cut cost. (1)

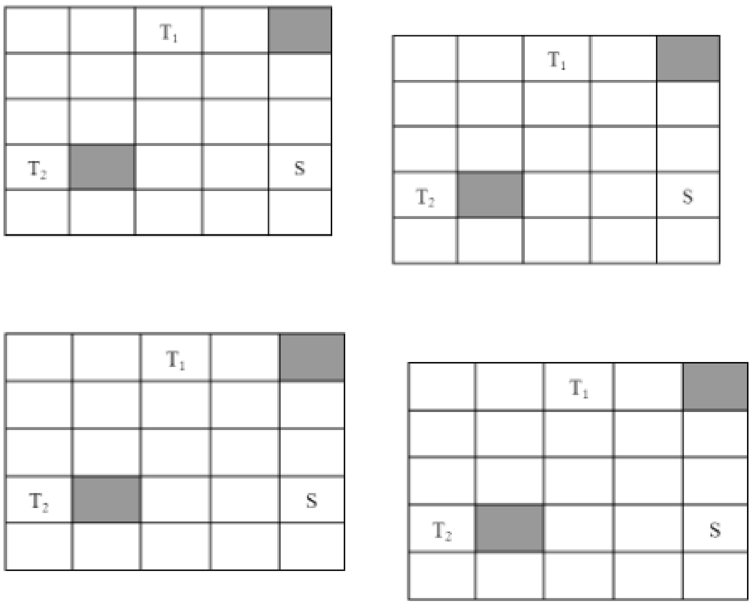
b. How many iterations will there be in the first pass? (1)

c. Perform the first pass of the KL algorithm. Identify the optimized partition. (7)

d. Are any further passes necessary? State the reason for your answer. (1)

Question-2

a. Use Lee's Maze algorithm to find the shortest path between S and the Ts avoiding obstacles. Dark regions are obstacles or components. **Show every iteration in separate squares. Use consecutive numbers to denote the grids in every iteration.** (6)



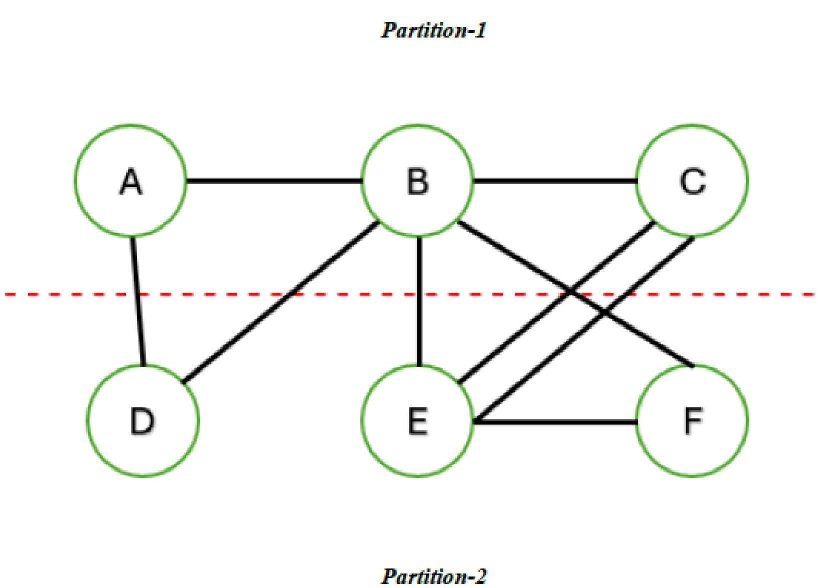
b. What is the memory requirement? (2)

c. What would have been the memory requirement if we denoted the grids as "1,2,3,1,2,3,....." (1)

d. What would have been the memory requirement if we denoted the grids as "0,0,1,1,0,0,1,1,....." (1)

Question-3

Optimize the following graph with KL Algorithm. Assume all connections have same weight. Answer the same Questions as seen in *Question-1.*



Question-4

For question (c), you are given a chip layout where you have 2 distinct sources and 2 distinct targets. You have to reach T1 from S1 and T2 from S2. The Lee Maze algorithm is an algorithm for signal routing, which uses a sequential approach.

Here, X is an obstacle for signal routing.

				X			S1		
			T2						
				X					
		T1				S2			

				X			S1		
			T2						
				X					
		T1				S2			

				X			S1		
			T2						
				X					
		T1				S2			

				X			S1		
			T2						
				X					
		T1				S2			

Using the Lee Maze algorithm, find the **shortest path** from S1 to T1 and from S2 to T2. Is there any chance for **multiple answers**? If so, which one is **better**? **Justify** your statement. [**Hints**: Refer to Lee Maze as a sequential approach]
[**N.B**- Use the top two and bottom two boxes for the different cases. Complete all iterations for each source in one single box, and then back propagate and dictate the new metal connection in that box]